WORKING SCIENTIFICALLY

Science Detectives: Working Scientifically

Observing, asking and predicting, fair tests, measuring, tables and graphs, patterns and conclusions


magnifying glass


test tube


ruler


stopwatch


bar chart

In this pack

Learn it: How to be a science detective ★☆☆
Fact check: Tools and steps ★☆☆
Think it through: Is it fair? ★★☆
Investigate: Does a higher drop make a higher bounce? ★★☆
Real-world case study: Dr Firdausi Qadri: science that saves lives ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · W1 · WORKING SCIENTIFICALLY

How to be a science detective

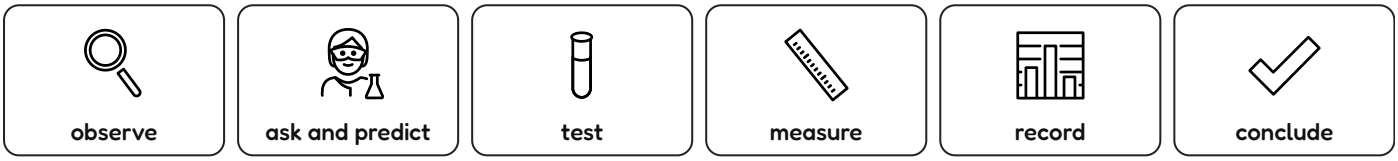
★☆☆ Grade 4

Name: Class: Date:

Learning goals

- Ask questions that can be tested and make predictions.
- Plan a fair test and measure carefully with the right tools and units.
- Present results in tables and graphs, find patterns and draw conclusions.

Detectives look for clues, collect evidence and use it to solve mysteries. Scientists do the same! They **observe** the world, ask **questions**, **test** their ideas and use **evidence** to find answers. You can work like a scientist too.



1. Observe and ask

Use your **senses** (and tools such as a hand lens) to look closely. Then ask a question you can **test**, such as "Which paper towel soaks up the most water?" or "Does a ball bounce higher if it is dropped from higher up?"

2. Predict

A **prediction** says what you think will happen **and why**: "I think the ball will bounce higher from higher up, **because** it will be moving faster when it hits the ground."

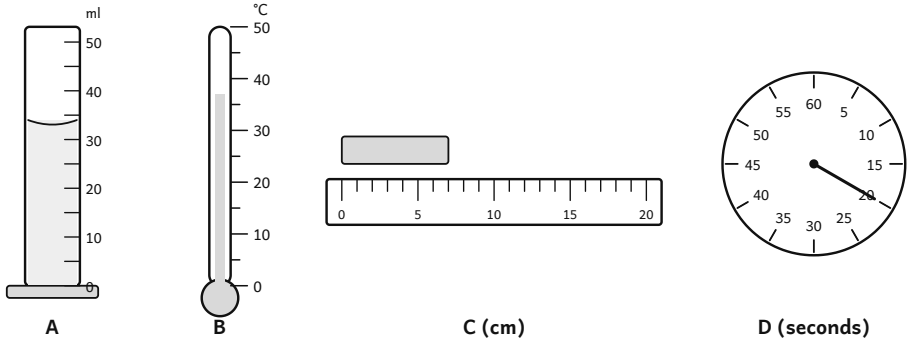
3. Plan a fair test

In a **fair test**, you change only **one** thing and keep everything else the same. Otherwise you will not know what caused the result.

I will change…	one thing (for example, the height of the drop)
I will measure…	the result (how high it bounces)
I will keep the same…	everything else (the same ball, the same floor, the same person measuring)

4. Measure carefully

To measure...	Use a...	Unit
length	ruler or tape measure	cm, m
volume	measuring cylinder or jug	ml, l
temperature	thermometer	°C
time	stopwatch	seconds (s), minutes
force	forcemeter	newtons (N)
mass	balance (scales)	g, kg



Read the scale with your eye level with the mark.

Do each test **more than once**. If the results are close, you can trust them more. If one result is very different, check it and measure again.

5. Record and present

Record results in a **table** with clear headings and units. Then show them in a **bar chart** (when you compare separate things, like different balls) or a **line graph** (when you measure something changing, like temperature over time).

6. Find patterns and conclude

Look for a **pattern**: "The higher the drop, the higher the bounce." Write a **conclusion** that answers your question, and say whether the results **support your prediction**. Do you have **enough** evidence, or should you test more?

7. Models and explaining

Scientists use **models** (drawings, globes, diagrams) to explain things that are too big, too small or too slow to see. They share their findings so others can check them, and they combine evidence with **creative ideas**.

Bangladesh connection

Dr Firdausi Qadri is a Bangladeshi scientist at icddr,b in Dhaka. Through careful tests with thousands of people, she helped bring low-cost vaccines against cholera and typhoid to the children of Bangladesh.

Key words

observe	look closely using your senses
prediction	what you think will happen, and why
fair test	change one thing; keep everything else the same
evidence	facts and measurements that support an idea
pattern	something that is repeated in the results
conclusion	what the results show
reliable	results you can trust because you repeated them
model	a way of showing an idea we cannot easily see

Tools and steps

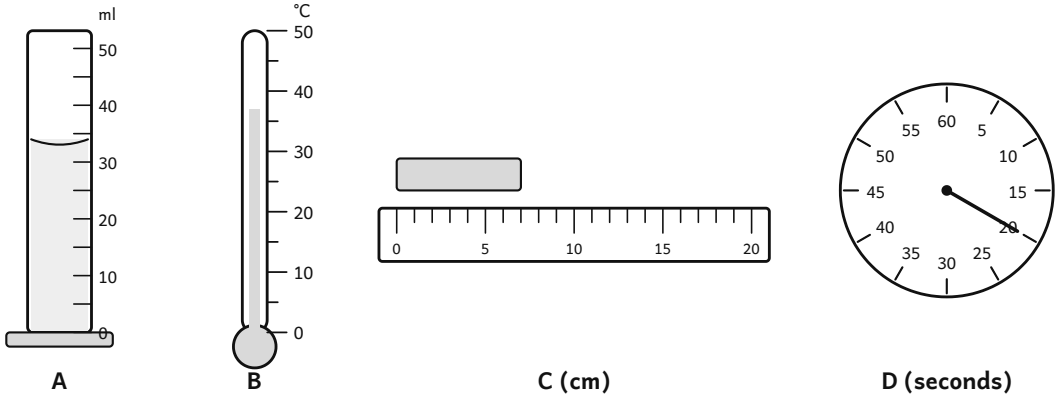
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Grade 4

Name: Class: Date:

Learning goals

- Choose equipment and units.
- Order the steps of an investigation.

1 Read each instrument. Include the unit. [4]



A B C (cm) D (seconds)

A	B	C	D

2 Match the measurement to the tool. [5]

Draw a line to match each pair, or write the letter.

how hot the water is ●	● **A** ruler
how long a race takes ●	● **B** measuring jug
how long a leaf is ●	● **C** forcemeter
how much juice is in a glass ●	● **D** stopwatch
how hard you pull a rope ●	● **E** thermometer

3 **Put the steps of an investigation in order.** [4]

- ☐ record the results
- ☐ ask a question
- ☐ write a conclusion
- ☐ do the test and measure
- ☐ make a prediction
- ☐ plan a fair test

4 **Bar chart or line graph?** [3]

Results	Best graph
favourite fruits of our class	
temperature of the classroom every hour	
how far four kinds of balls bounce	

Is it fair?

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Spot unfair tests.
- Write predictions.
- Find patterns.

1 **Sami wants to find out which paper towel soaks up the most water. He uses a big piece of towel A and a small piece of towel B, and dips them for different times. Is this fair? Explain how to make it fair.** [3]

2 **Write a prediction with a reason: "Will an ice cube melt faster in the sun or in the shade?"** [2]

3 **Look at the results. What is the pattern?** [2]

Amount of water each day (ml)	0	25	50	100
Height of bean plant after 2 weeks (cm)	2	9	15	16

4 **Three children timed the same race: 12 s, 13 s and 25 s. Which result is odd? What should they do?** [2]

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Does a higher drop make a higher bounce?

★★☆ Grade 4

Name: Class: Date:

Learning goals

- Plan, measure and repeat.
- Draw a line graph.
- Conclude and evaluate.

Class 4A dropped a tennis ball from different heights onto the same floor and measured how high it bounced. They did each drop three times.

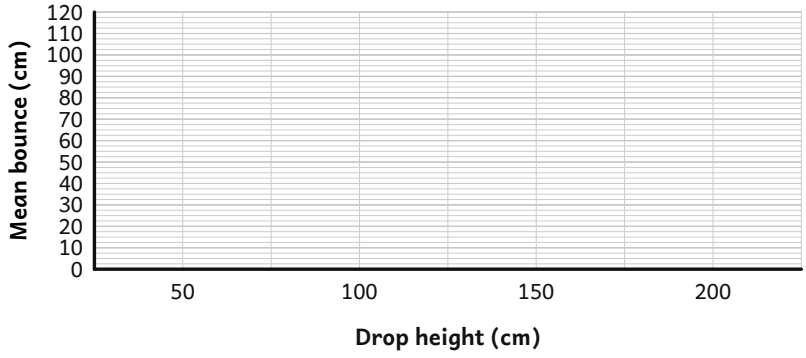
1 Complete the plan. [4]

Question	
Change	
Measure	
Keep the same	
Prediction	

2 Complete the means. [4]

Drop height (cm)	Bounce 1	Bounce 2	Bounce 3	Mean bounce (cm)
50	28	30	29	
100	57	60	57	
150	86	88	90	
200	115	118	115	

3 Draw a line graph of the means. [3]



4 **Write *a* conclusion. Did the results support the prediction?** [2]

.....

.....

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Dr Firdausi Qadri: science that saves lives

★★★ Grade 4

Name: Class: Date:

Learning goals

- Learn how a real scientist uses evidence.
- Read data from a trial.
- Think about how science helps people.

Science · Health · Maths · Citizenship

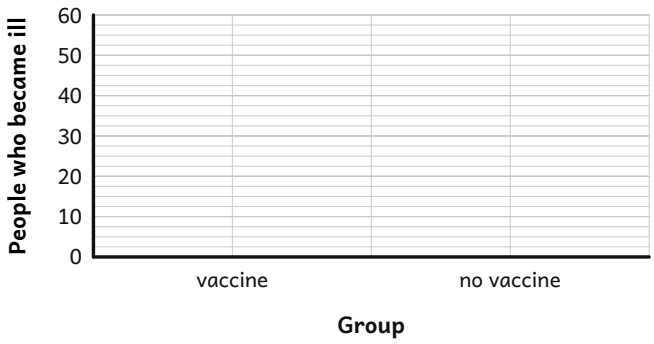
Cholera and **typhoid** are diseases spread by dirty water. They can make children very ill. **Dr Firdausi Qadri**, a scientist at **icddr,b** in Dhaka, has spent her life finding ways to protect people from them.

Scientists cannot just guess that a vaccine works; they need **evidence**. So Dr Qadri and her team ran large **trials** with thousands of volunteers in Dhaka. Some people got the vaccine and others did not, and the team counted how many in each group became ill. The evidence helped make cheap vaccines available to millions of children. In 2021 she won the Ramon Magsaysay Award, one of Asia's top prizes, for her work.

Based on information about Dr Firdausi Qadri and icddr,b. Numbers below are example data.

1 **A vaccine trial (example data): 1000 people in each group.** [3]

Group	Got the vaccine	Did not get the vaccine
People who became ill	10	50



a) Draw the bar chart.

b) How many times more people became ill without the vaccine?

.....

2 **Why did the trial need two groups?** [2]

.....

.....

3 **Why was it important to test thousands of people, not just ten?** [2]

4 **Besides vaccines, how can people protect themselves from diseases spread by dirty water?** [2]

Show what you know

★★★ Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

M	O	L	M	R	D	F	A	I	R
T	E	V	E	O	H	C	S	E	M
N	V	P	A	B	V	F	P	R	F
S	I	A	S	S	T	E	R	E	H
J	D	T	U	E	E	P	E	S	G
U	E	T	R	R	S	G	D	U	R
J	N	E	E	V	T	O	I	L	A
L	C	R	E	E	M	J	C	T	P
B	E	N	A	O	W	D	T	O	H
D	H	O	K	S	L	H	F	F	B

evidence fair graph measure observe pattern predict result test

2 In a fair test you change... [1]

◯ everything ◯ only one thing

◯ two things ◯ nothing

3 Which tool measures temperature? [1]

◯ ruler ◯ stopwatch

◯ thermometer ◯ forcemeter

4 Plan your own investigation: "Which ball bounces highest?" Write what you will change, measure and keep the same. [3]

Change	
Measure	
Keep the same	

5 **Challenge. Why do scientists share their results with other scientists?** [2]

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
ask a question I can test	○	○	○
write a prediction with a reason	○	○	○
plan a fair test	○	○	○
measure with the right tool and unit	○	○	○
draw a bar chart or line graph	○	○	○
write a conclusion using evidence	○	○	○